

Generation Interconnection System Impact Study Report

For Queue Project Q13-0216

Puddledock Rd 138 kV

August 16, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0216. The Interconnection Customer proposes a 70.0 MW battery energy storage facility interconnecting at the Puddledock Rd 138 kV substation in the Virginia area. The request was received on March 24, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 154 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 47 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$75,706,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The CARODEAN-REDLAND 138 kV is overloaded to 179.5% of its 222 MVA emergency rating for the loss of the DAYTON-CRANDALL 345 kV. The project contributes 5.9% DFAX (4.1 MW) to this overload.

GRENBANK-WEYERS CAVE 138 kV: 160.4% loading (265 MVA) following the outage of the CRANDALL SUBSTATION-DAYTON 345 kV Line; DFAX = 8.0%.

GROTTOES-WEYERS CAVE 138 kV: 160.3% loading (258 MVA) following the outage of the CRANDALL-DAYTON 345 kV; DFAX = 8.0%.

The HARRISONBURG-SANS SOUCI 138 kV is overloaded to 154.9% of its 254 MVA emergency rating for the loss of the DAYTON-CHATOM 345 kV. The project contributes 12.7% DFAX (8.9 MW) to this overload.

The OAK CRK STA-WADSWORTH 138 kV is overloaded to 154.7% of its 230 MVA emergency rating for the loss of the CHATOM-DAYTON 345 kV L. The project contributes 12.7% DFAX (8.9 MW) to this overload.

For the loss of DAYTON-CHATOM 345 kV, the SANS SOUCI SUBSTATION-OAK CRK STATION 138 kV loads to 154.4% of its 255 MVA rating. Project DFAX: 12.7%.

NICHOLS-ST LOUIS 138 kV: 151.9% loading (185 MVA) following the outage of the PAGE-PURDY SUBSTATION 138 kV; DFAX = 17.9%.

The CROCKETT-LORETTO 138 kV line is overloaded to 146.9% of its 380 MVA emergency rating for the loss of the AMERICAN-URBAN 345 kV Line. The project contributes 5.9% DFAX (4.2 MW) to this overload.

S BELLEVILLE-NICHOLS 138 kV: 140.3% loading (197 MVA) following the outage of the PAGE-PURDY 138 kV; DFAX = 17.9%.

TES ILER CITY STA-BRIDGE RIVER 138 kV: 135.7% loading (206 MVA) following the outage of the AMERICAN-URBAN 345 kV line; DFAX = 12.6%.

The INNOVATIVE SOLAR 31-KAPSTONE 138 kV is overloaded to 132.2% of its 243 MVA emergency rating for the loss of the LOUISA GENERATION FACILITY-GRIMES 138 kV L. The project contributes 9.9% DFAX (6.9 MW) to this overload.

INNOVATIVE SOLAR 31-PRATT'S 138 kV: 131.5% loading (240 MVA) following the outage of the LOUISA GENERATION FACILITY-GRIMES 138 kV; DFAX = 9.9%.

For the loss of PURDY-CHATOM 138 kV, the DPC PRAIRIE DU CHIEN SUBSTATION-FRITO LAY 138 kV loads to 131.1% of its 233 MVA rating. Project DFAX: 38.9%.

S BELLEVILLE-NORTHPOINT 138 kV: 129.7% loading (208 MVA) following the outage of the PAGE-PURDY 138 kV; DFAX = 17.9%.

For the loss of ASPEN-CRANDALL SUBSTATION 345 kV, the SALAMANCAROCHESTER ST-DECLO 138 kV line loads to 128.5% of its 259 MVA rating. Project DFAX: 8.3%.

VERONA-REDLAND 138 kV: 128.0% loading (238 MVA) following the outage of the CRANDALL-DAYTON SUBSTATION 345 kV L; DFAX = 5.2%.

For the loss of URBAN-AMERICAN 345 kV Line, the BRIDGE RIV-ABBOTSFORD 138 kV loads to 127.1% of its 210 MVA rating. Project DFAX: 6.9%.

For the loss of DAYTON-CRANDALL SUBSTATION 345 kV, the VERONA-GREENBANK 138 kV L loads to 126.9% of its 234 MVA rating. Project DFAX: 5.2%.

For the loss of OAK CREEK STATION-SANS SOUCI 138 kV, the NUCLEAR FUELS-ALVIN 138 kV loads to 126.6% of its 201 MVA rating. Project DFAX: 14.0%.

PAGE-PURDY SUBSTATION 138 kV: 126.3% loading (261 MVA) following the outage of the ST LOUIS-NICHOLS 138 kV; DFAX = 23.9%.

DETOUR-CEDARVILLE JUNCTION 138 kV L: 126.2% loading (248 MVA) following the outage of the ST LOUIS-NICHOLS 138 kV; DFAX = 23.9%.

CEDARVILLE JUNCTION-PAGE 138 kV: 126.0% loading (252 MVA) following the outage of the ST LOUIS-NICHOLS 138 kV L; DFAX = 23.9%.

The CHATOM-ALVIN 138 kV is overloaded to 125.3% of its 208 MVA emergency rating for the loss of the OAK CRK STA-SANS SOUCI 138 kV line. The project contributes 14.0% DFAX (9.8 MW) to this overload.

The CHERRY-MOUNT JACKSON 138 kV is overloaded to 124.5% of its 178 MVA emergency rating for the loss of the PURDY-CHATOM 138 kV Line. The project contributes 64.3% DFAX (45.0 MW) to this overload.

WESTCHASE-COFFEEN N 138 kV: 124.4% loading (346 MVA) following the outage of the CRANDALL-DAYTON 345 kV; DFAX = 24.9%.

For the loss of CRANDALL-DAYTON 345 kV, the HARRISONBURG-COFFEEN N 138 kV loads to 124.4% of its 346 MVA rating. Project DFAX: 24.9%.

STATION 54-WESTCHASE 138 kV: 122.1% loading (369 MVA) following the outage of the DAYTON-CRANDALL 345 kV; DFAX = 24.9%.

The CROZET-REVITAL 138 kV is overloaded to 121.1% of its 167 MVA emergency rating for the loss of the PAGE-PURDY 138 kV. The project contributes 6.2% DFAX (4.4 MW) to this overload.

The CHERRY-DPC PRAIRIE DU CHIEN 138 kV line is overloaded to 120.2% of its 192 MVA emergency rating for the loss of the PURDY-CHATOM 138 kV. The project contributes 64.3% DFAX (45.0 MW) to this overload.

NUCLEAR FUELS-ST LOUIS 138 kV: 117.3% loading (192 MVA) following the outage of the WADSWORTH-ROOSEVELT RD 138 kV; DFAX = 12.6%.

For the loss of CRANDALL-DAYTON 345 kV, the CARODEAN-CRANDALL 138 kV loads to 116.6% of its 365 MVA rating. Project DFAX: 5.9%.

NORTHPOINT-MITCHELL GARDENS 138 kV line: 113.6% loading (222 MVA) following the outage of the PAGE-PURDY 138 kV; DFAX = 17.9%.

TOONE SUBSTATION-N OMAHA 138 kV: 112.8% loading (255 MVA) following the outage of the CRANDALL-ASPEN 345 kV line; DFAX = 11.1%.

The STA 54-TOONE 138 kV L is overloaded to 112.2% of its 248 MVA emergency rating for the loss of the ASPEN-CRANDALL 345 kV Line. The project contributes 11.1% DFAX (7.8 MW) to this overload.

For the loss of PAGE-PURDY 138 kV, the DALE MABRY SUBSTATION-REVITAL 138 kV line loads to 107.5% of its 193 MVA rating. Project DFAX: 6.2%.

CRANDALL SUBSTATION 345-138 kV Auto: 107.5% loading (457 MVA) following the outage of the CRANDALL-DAYTON 345 kV; DFAX = 5.1%.

For the loss of PURDY-PAGE SUBSTATION 138 kV Line, the MITCHELL GARDENS-KAPSTONE 138 kV loads to 107.0% of its 227 MVA rating. Project DFAX: 17.9%.

For the loss of CRANDALL-ASPEN 345 kV line, the CARD-YUCAIPA 138 kV line loads to 106.6% of its 267 MVA rating. Project DFAX: 7.0%.

DETOUR-SILOAM SPRINGS 138 kV L: 105.8% loading (181 MVA) following the outage of the DETUR SUBSTATION-FORT BENTON 138 kV L; DFAX = 16.7%.

The CARD-CRAGGY 138 kV Line is overloaded to 105.6% of its 258 MVA emergency rating for the loss of the CRANDALL-ASPEN 345 kV Line. The project contributes 7.0% DFAX (4.9 MW) to this overload.

HARRISONBURG-MOUNT CLINTON 138 kV L: 104.4% loading (267 MVA) following the outage of the CHATOM-DAYTON 345 kV; DFAX = 19.2%.

The GROTTUES-DALE MABRY 138 kV is overloaded to 104.1% of its 215 MVA emergency rating for the loss of the PAGE-PURDY SUBSTATION 138 kV L. The project contributes 6.2% DFAX (4.4 MW) to this overload.

For the loss of NUCLEAR FUELS-ST LOUIS SUBSTATION 138 kV, the ROOSEVELT RD-ELKTON D.P 138 kV L loads to 104.1% of its 197 MVA rating. Project DFAX: 10.4%.

The ELKTON D.P-HOUSTON 138 kV is overloaded to 103.1% of its 199 MVA emergency rating for the loss of the ST LOUIS-NUCLEAR FUELS 138 kV. The project contributes 10.4% DFAX (7.2 MW) to this overload.

STUARTS DRAFT-CRAGGY 138 kV L: 102.9% loading (239 MVA) following the outage of the CRANDALL-ASPEN 345 kV Line; DFAX = 7.0%.

For the loss of NUCLEAR FUELS SUBSTATION-ST LOUIS 138 kV Line, the ROOSEVELT RD-WADSWORTH 138 kV line loads to 102.6% of its 208 MVA rating. Project DFAX: 10.4%.

For the loss of DAYTON-CHATOM 345 kV, the MOUNT CLINTON-FOREST JCT 138 kV loads to 102.3% of its 253 MVA rating. Project DFAX: 19.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CARODEAN-REDLAND 138 kV	Rebuild line with higher-capacity conductor	\$1,268,000	3% (\$41,000)
GREENBANK-WEYERS CAVE 138 kV line	Rebuild line with higher-capacity conductor	\$2,475,000	5% (\$119,000)
GROTTUES-WEYERS CAVE 138 kV	Rebuild line with higher-capacity conductor	\$6,871,000	5% (\$332,000)
SANS SOUCI-HARRISONBURG 138 kV	Rebuild line with higher-capacity conductor	\$3,420,000	11% (\$384,000)
OAK CRK STA-WADSWRTH 138 kV	Rebuild line with higher-capacity conductor	\$7,439,000	11% (\$835,000)
SANS SOUCI-OAK CRK STATION 138 kV	Rebuild line with higher-capacity conductor	\$3,008,000	11% (\$338,000)
ST LOUIS SUBSTATION-NICHOLS 138 kV	Rebuild line with higher-capacity conductor	\$7,067,000	16% (\$1,116,000)
LORETTO-CROCETT 138 kV L	Rebuild line with higher-capacity conductor	\$5,001,000	6% (\$303,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S BELLEVILLE-NICHOLS 138 kV Line	Rebuild line with higher-capacity conductor	\$4,776,000	16% (\$754,000)
TES ILER CITY STA-BRIDGE RIVER 138 kV	Rebuild line with higher-capacity conductor	\$7,644,000	47% (\$3,570,000)
KAPSTONE-INNOVATIVE SOLAR 31 138 kV	Rebuild line with higher-capacity conductor	\$8,058,000	58% (\$4,678,000)
INNOVATIVE SOLAR 31-PRATT'S 138 kV	Rebuild line with higher-capacity conductor	\$6,832,000	58% (\$3,967,000)
FRITO LAY-DPC PRAIRIE DU CHIEN 138 kV L	Rebuild line with higher-capacity conductor	\$4,593,000	27% (\$1,246,000)
S BELLEVILLE-NORTHPOINT 138 kV	Rebuild line with higher-capacity conductor	\$4,970,000	16% (\$785,000)
SALAMANCAROCHESTER ST-DECLO 138 kV line	Rebuild line with higher-capacity conductor	\$33,393,000	34% (\$11,388,000)
REDLAND SUBSTATION-VERONA 138 kV	Rebuild line with higher-capacity conductor	\$6,420,000	4% (\$277,000)
BRIDGE RIV-ABBOTSFORD 138 kV	Rebuild line with higher-capacity conductor	\$18,510,000	27% (\$5,057,000)
VERONA-GREENBANK 138 kV L	Rebuild line with higher-capacity conductor	\$3,716,000	4% (\$161,000)
NUCLEAR FUELS-ALVIN 138 kV	Rebuild line with higher-capacity conductor	\$7,046,000	18% (\$1,283,000)
PAGE-PURDY SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$3,957,000	21% (\$835,000)
CEDARVILLE JUNCTION SUBSTATION-DETOUR 138 kV	Rebuild line with higher-capacity conductor	\$10,001,000	21% (\$2,111,000)
PAGE SUBSTATION-CEDARVILLE JCT 138 kV	Rebuild line with higher-capacity conductor	\$7,371,000	21% (\$1,556,000)
CHTOM-ALVIN 138 kV	Rebuild line with higher-capacity conductor	\$3,277,000	18% (\$597,000)
CHERRY-MOUNT JACKSON 138 kV	Rebuild line with higher-capacity conductor	\$9,191,000	100% (\$9,191,000)
WESTCHASE-COFFEEN N 138 kV line	Rebuild line with higher-capacity conductor	\$3,886,000	21% (\$830,000)
HARRISONBURG-COFFEEN N 138 kV	Rebuild line with higher-capacity conductor	\$1,810,000	21% (\$387,000)
STATION 54-WESTCHASE 138 kV	Rebuild line with higher-capacity conductor	\$6,730,000	21% (\$1,437,000)
CROZET-REVITAL 138 kV	Rebuild line with higher-capacity conductor	\$4,985,000	7% (\$346,000)
CHERRY-DPC PRAIRIE DU CHIEN 138 kV	Rebuild line with higher-capacity conductor	\$5,439,000	100% (\$5,439,000)
ST LOUIS-NUCLEAR FUELS 138 kV Line	Rebuild line with higher-capacity conductor	\$9,034,000	15% (\$1,338,000)
CARODEAN-CRANDALL 138 kV	Rebuild line with higher-capacity conductor	\$4,659,000	3% (\$152,000)
NORTHPOINT-MITCHELL GARDENS 138 kV line	Rebuild line with higher-capacity conductor	\$5,645,000	16% (\$891,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N OMAHA-TOONE 138 kV	Rebuild line with higher-capacity conductor	\$8,435,000	31% (\$2,637,000)
STA 54-TOONE 138 kV L	Rebuild line with higher-capacity conductor	\$2,228,000	31% (\$696,000)
REVITAL-DALE MABRY 138 kV	Reconductor line	\$1,812,000	7% (\$126,000)
CRANDALL SUBSTATION 345-138 kV Auto	Replace transformer with higher-rated unit	\$27,749,000	2% (\$687,000)
MITCHELL GARDENS-KAPSTONE 138 kV line	Reconductor line	\$1,881,000	16% (\$297,000)
CARD-YUCAIPA 138 kV line	Reconductor line	\$1,355,000	44% (\$594,000)
DETOUR-SILOAM SPRINGS 138 kV L	Reconductor line	\$6,603,000	74% (\$4,888,000)
CARD-CRAGGY 138 kV Line	Reconductor line	\$3,838,000	44% (\$1,682,000)
HARRISONBURG-MOUNT CLINTON 138 kV L	Reconductor line	\$1,848,000	22% (\$411,000)
DALE MABRY-GROTTOES 138 kV	Reconductor line	\$1,799,000	7% (\$125,000)
ROOSEVELT RD-ELKTON D.P 138 kV L	Reconductor line	\$445,000	13% (\$58,000)
ELKTON D.P-HOUSTON 138 kV	Reconductor line	\$651,000	13% (\$85,000)
STUARTS DRAFT-CRAGGY 138 kV L	Reconductor line	\$1,965,000	44% (\$861,000)
ROOSEVELT RD-WADSWORTH 138 kV line	Reconductor line	\$2,006,000	13% (\$263,000)
MOUNT CLINTON-FOREST JCT 138 kV	Reconductor line	\$2,481,000	22% (\$552,000)
	Total allocated to project		\$75,706,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.